G14 Hospital Database Management System

Faculty: Applied and Computer Sciences
Department: Computer Sciences
Subject: Information System 1.2
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Group: 14
Assessment 3: Project Plan



Group 14

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Agenda

1. The Scenario and Introduction
2. The Solution
3. Business Rules
4. Database Normalization
5. ERD Phase
6. SQL Code
7. SQL Query Results
8. Conclusion

The Scenario

1

A patient arrives at G14 Healthcare Facility in Vanderbijlpark, requiring medical attention

2

The patient's medical record must be located before care can proceed

3

Physical files and manual record-keeping make finding the correct file slow

4

Other patients keep arriving, queues grow and staff face increasing pressure

The challenge

Delays in retrieving medical records can affect the patient experience and make it difficult for healthcare workers to access important information when they need it.

How records are kept

Patient information, appointments, treatments, prescriptions and billing details are stored in physical documents and separate files.

Introduction

Who we serve	G14 Healthcare Facility provides basic healthcare services to the Vanderbijlpark community.
What operations rely on	Accurate and accessible information for patient care, appointment scheduling, treatment management, prescription records, billing and staff administration.
The challenge	The existing flat-file system is becoming less suitable as the patient population grows. It relies heavily on manual processes, making information difficult to retrieve, update and maintain consistently.

Introduction: Key Problems Identified

Slow information retrieval

Staff must manually search through physical files, delaying access to patient information.

Poor record storage

Paper documents occupy significant space and can be lost, damaged or misplaced.

Data duplication

The same patient information may be recorded in appointment, treatment, prescription and billing files.

Difficult information updates

Changes to patient details may need to be made across several files, creating opportunities for human error.

Data integrity and security concerns

Complete, accurate and protected records are harder to maintain when information is scattered.

Inefficient reporting

Compiling reports and analysing operations is challenging when information is in separate files.

The Solution

Replace G14's existing flat-file system with a relational database management system

Group 14 proposes a structured database solution designed around G14's information management needs.

Structured, related tables

Information is organised into related tables rather than scattered across separate physical files.

Primary and foreign keys

Keys establish relationships between the different types of records.

Linked, not duplicated

A patient's information links to appointments, treatments and prescriptions without repeating details in every record.

The Solution: How It Works

Proposed tables

Patients, doctors, appointments, treatments, prescriptions, bills and staff members. Each table stores information relevant to its purpose.

Database normalisation

Progressing from Unnormalized Form (UNF) through 1NF and 2NF to 3NF reduces data duplication, improves consistency and maintains data integrity.

Security and backup

Appropriate access controls, security measures and backup procedures protect sensitive patient information from loss or unauthorised access.

Expected outcome

Reduced administrative burden, better record organisation and better access to patient information. The database cannot independently guarantee shorter waiting times or better clinical outcomes.

Business Rules

The rules describe how G14 Healthcare Facility operates and how data is managed within the proposed relational database, supporting day-to-day hospital operations.

Patient BR1 to BR5	**Staff** BR6 to BR11	**Appointment** BR12 to BR19	**Admission** BR20 to BR26	**Ward** BR27 to BR30
Room BR31 to BR39	**Treatment** BR40 to BR46	**Medicine** BR47 to BR52	**Prescription** BR53 to BR60	**Billing** BR61 to BR65

Patient and Staff Rules

Patient Rules

- **BR1** Each patient is identified by a PatientID
- **BR2** A patient can have one or more appointments
- **BR3** Each appointment has exactly one patient
- **BR4** A patient can have one or more billing records
- **BR5** A patient can be in the system without an appointment

Staff Rules

- **BR6** Every staff member has their own StaffID
- **BR7** Every staff member has a StaffType: Doctor, Nurse, Receptionist or Administrator
- **BR9** A staff member can be in multiple treatments
- **BR10** Only Doctors and Nurses can be assigned to treatments
- **BR11** Receptionists and Administrators cannot give or supervise treatments

Appointment and Admission Rules

Appointment Rules

- **BR12** Each appointment has a unique AppointmentID
- **BR13** Each appointment has one patient assigned
- **BR15** The patient must be in the system before an appointment is created
- **BR17** An appointment can yield 0 or 1 admissions
- **BR18** Zero, one or many prescriptions per appointment
- **BR19** Appointment status is recorded

Admission Rules

- **BR20** Each admission has a unique AdmissionID
- **BR21** Every admission comes from one appointment
- **BR22** Each admission is given a room
- **BR24** No admission without an appointment and room
- **BR25** Admission and discharge dates are documented
- **BR26** A patient can have several admissions over time

Ward and Room Rules

Ward Rules

- **BR27** Every ward has a WardID
- **BR28** A ward can have one or more rooms
- **BR29** Each room is assigned to only one ward

Room Rules

- **BR31** Every room has a RoomID
- **BR33** A room can serve several admissions over time
- **BR34** One patient per room at a given time
- **BR36** RoomAvailability is Available or Occupied
- **BR37** Patients can only be put in an Available room
- **BR38** On admission the room becomes Occupied
- **BR39** On discharge the room becomes Available

Treatment and Medicine Rules

Treatment Rules

- **BR40** Each treatment has a unique TreatmentID
- **BR41** All treatments belong to an appointment
- **BR42** Each treatment is carried out or supervised by one staff member
- **BR43** One or more treatments per appointment
- **BR44** Diagnosis and treatment description are recorded
- **BR46** Treatment can be for an outpatient or an admitted patient

Medicine Rules

- **BR47** Each medicine has a unique MedicineID
- **BR48** Every medicine has a MedicineName
- **BR49** Every medicine has a stock value
- **BR50** A medicine can be on several prescriptions
- **BR51** A medicine can exist without a prescription
- **BR52** Stock is adjusted when medication is supplied

Prescription and Billing Rules

Prescription Rules

- **BR53** Each prescription has a unique PrescriptionID
- **BR54** Each prescription is linked to an appointment
- **BR55** Every prescription includes one medicine
- **BR58** Multiple prescriptions can be issued for one appointment
- **BR60** Dosage is recorded for each prescription

Billing Rules

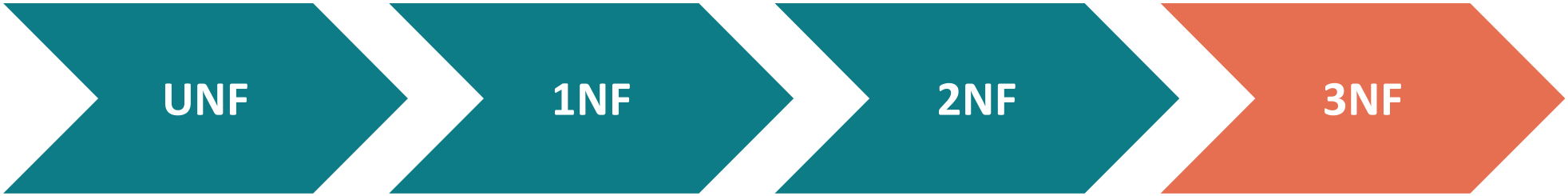
- **BR61** Each bill has a unique BillID
- **BR62** Each bill belongs to one patient
- **BR63** A patient may have multiple billing records
- **BR64** PaymentStatus is Paid, Pending or Overdue
- **BR65** Bills reflect services provided in appointments, admissions, treatments and prescriptions

Relationship and Cardinality Summary

One	Can have	Related records
Patient	many	Appointments
Patient	many	Billing records
Appointment	many	Treatments
Appointment	many	Prescriptions
Appointment	zero or one	Admission
Ward	several	Rooms
Room	many, over time	Admissions
Staff member	many	Treatments
Medicine	several	Prescriptions

Database Normalization: Introduction

- Normalization converts the existing flat-file system into a structured relational database
- The flat file combines patient, staff, treatment, room and billing data, causing duplicated records, inefficient retrieval and update challenges
- Performed step by step to reduce duplication, ensure data integrity and meet G14's operational requirements



Unnormalized Form to First to Second to Third Normal Form

Unnormalized Form (UNF)

PatientID	PatientName	AppointmentID	TreatmentID	PrescriptionID	MedicineName	WardName	BillAmount
P001	John Smith	A001	T001	PR001	Amlodipine	General	1500
P001	John Smith	A001	T002	PR002	Aspirin	General	1500
P002	Mary Jones	A002	T003	PR003	Paracetamol	Surgical	2200
P003	David Brown	A003	T004	PR004	Metformin	ICU	1800

Problems Identified

- Patient information is repeated
- Appointment information is repeated
- Medicine information is repeated
- Billing information appears multiple times
- Data redundancy exists throughout the table
- Updating, deleting or inserting data may cause anomalies

First Normal Form (1NF)

Patient

PatientID	FirstName	LastName	PhoneNumber
P001	John	Smith	0823456789
P002	Mary	Jones	0834567890
P003	David	Brown	0845678901

Appointment

AppointmentID	PatientID	Date	Time	Status
A001	P001	01/04/2026	08:00	Completed
A002	P002	03/04/2026	09:30	Completed
A003	P003	05/04/2026	10:00	Completed

Admission

AdmissionID	AppointmentID	PatientID	RoomID	Admitted	Discharged
AD001	A001	P001	R01	01/04/2026	05/04/2026
AD002	A002	P002	R02	03/04/2026	07/04/2026

1NF rule

All repeating groups are removed and each attribute is restricted to a single atomic value per entry.

1NF: Treatment, Medicine, Prescription

Treatment

TreatmentID	AppointmentID	StaffID	Diagnosis
T001	A001	S001	Hypertension
T002	A001	S002	Monitoring
T003	A002	S003	Influenza

Medicine

MedicineID	MedicineName	Stock
M001	Amlodipine	150
M002	Aspirin	300
M003	Paracetamol	500

Prescription

PrescriptionID	AppointmentID	MedicineID	Dosage
PR001	A001	M001	5mg Daily
PR002	A001	M002	100mg Daily
PR003	A002	M003	500mg Every 6 Hours

After 1NF

- Repeating groups were eliminated
- Atomic values were enforced
- Each row became uniquely identifiable
- Data became easier to manage

Second Normal Form (2NF)

All non-key attributes depend on the primary key.

Staff

StaffID	StaffType	Initials	PhoneNumber
S001	Doctor	JM	1236547890
S002	Nurse	PN	0174852369
S003	Doctor	LK	0321458796

Ward

WardID	WardName
W01	General
W02	Surgical
W03	ICU

Billing

BillID	PatientID	Amount	PaymentStatus
B001	P001	1500	Paid
B002	P002	2200	Pending
B003	P003	1800	Paid

Room

RoomID	WardID	WardName	RoomNumber	RoomAvailability
R01	W01	General	101	Occupied
R02	W02	Surgical	201	Occupied
R03	W01	General	102	Occupied
R04	W03	ICU	301	Occupied
R05	W02	Surgical	202	Occupied

Second Normal Form (2NF): Improvements

After 2NF

- Partial dependencies were removed
- Data was separated into subject-specific entities
- Medicine information was no longer duplicated
- Prescription management improved
- Data redundancy was significantly reduced

A problem still exists in the Room table

WardName depends on WardID rather than RoomID.
This transitive dependency violates 3NF.

This takes us to the next step: Third Normal Form.

Third Normal Form (3NF)

All transitive dependencies are removed.

Room

RoomID	WardID	RoomNumber	RoomAvailability
R01	W01	101	Occupied
R02	W02	201	Occupied
R03	W01	102	Occupied
R04	W03	301	Occupied
R05	W02	202	Occupied

Transitive dependency removed

WardName now lives only in the Ward table, so the Room table achieves 3NF.

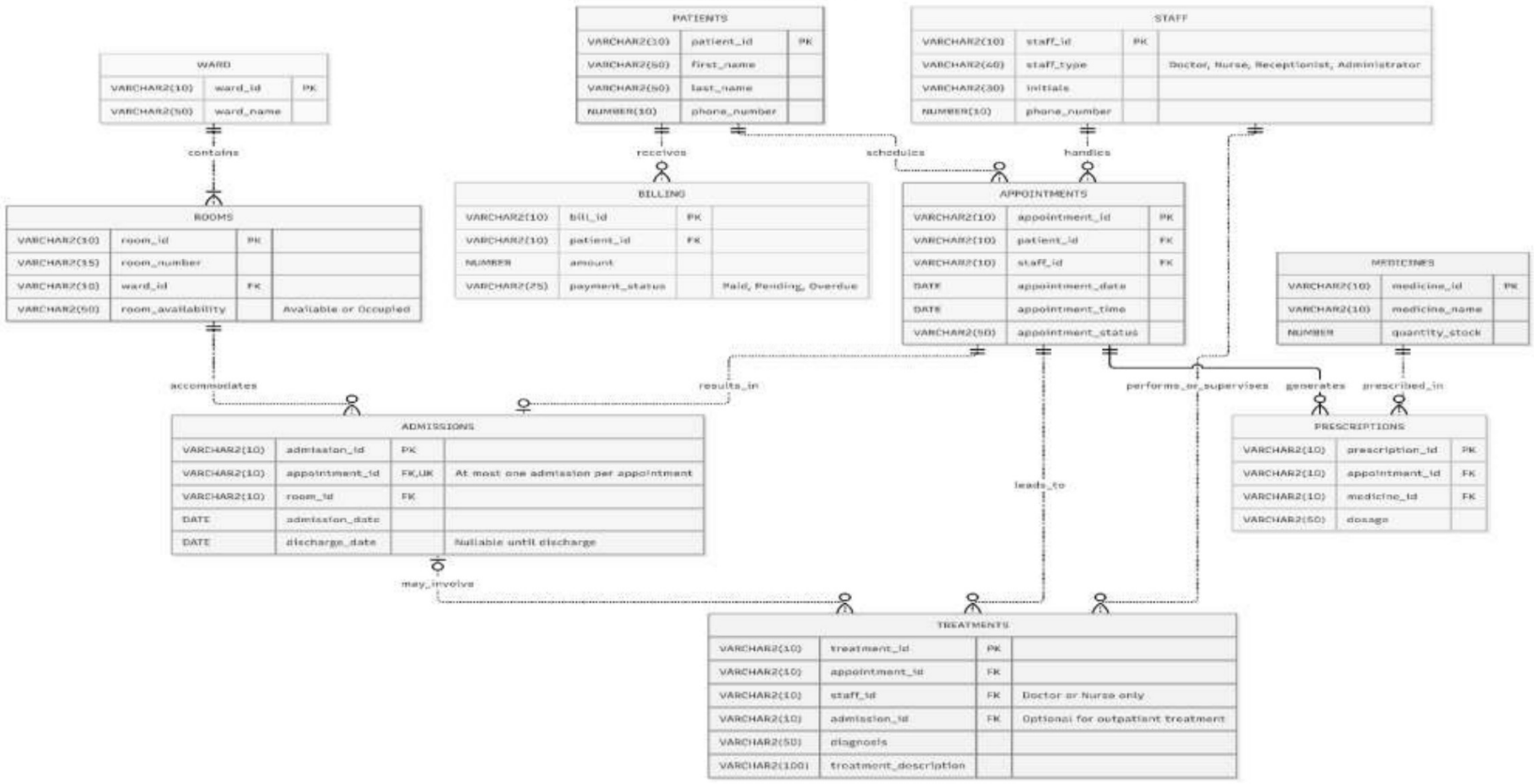
We achieve the following because of 3NF

- Repeating groups are eliminated
- Partial dependencies are removed
- Transitive dependencies are removed
- Medicine data is separated from prescription data
- Appointment information is stored once
- Foreign keys link related tables
- Data integrity is improved
- Redundancy is minimized throughout the database

Final 3NF Relations

Entities	Attributes
Patient	PatientID (PK), FirstName, LastName, PhoneNumber
Appointment	AppointmentID (PK), PatientID (FK), AppointmentDate, AppointmentTime, AppointmentStatus
Admission	AdmissionID (PK), AppointmentID (FK), PatientID (FK), RoomID (FK), AdmissionDate, DischargeDate
Ward	WardID (PK), WardName
Room	RoomID (PK), WardID (FK), RoomNumber, RoomAvailability
Staff	StaffID (PK), StaffType, Initials
Treatment	TreatmentID (PK), AppointmentID (FK), StaffID (FK), Diagnosis, TreatmentDescription
Medicine	MedicineID (PK), MedicineName, Stock
Prescription	PrescriptionID (PK), AppointmentID (FK), MedicineID (FK), Dosage
Billing	BillID (PK), PatientID (FK), Amount, PaymentStatus

The Resulting ERD



Database Normalization: Summary



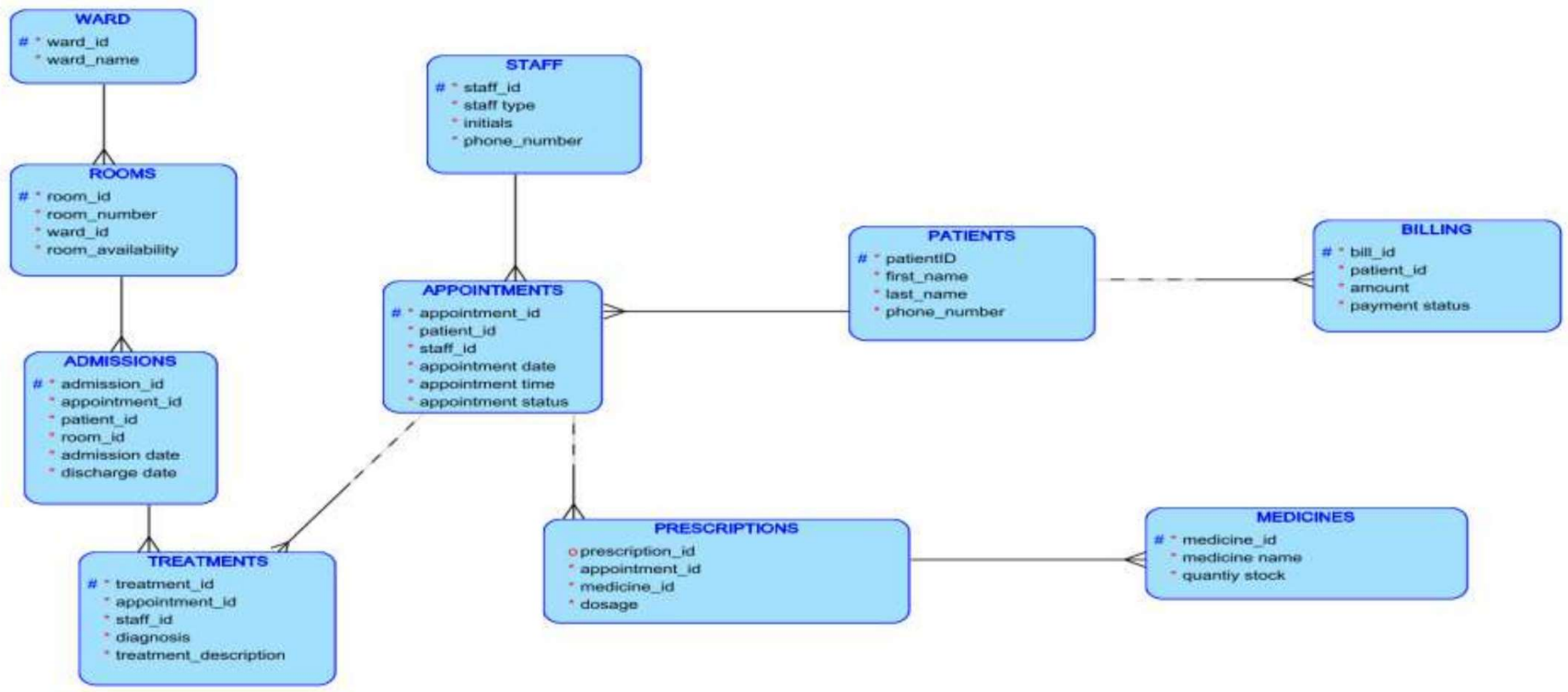
- Normalization eliminated redundant data, improved consistency and ensured efficient data storage
- The final structure manages patients, appointments, admissions, wards, rooms, staff, treatments, medicines, prescriptions and billing
- It supports both outpatient and inpatient healthcare services

Next step

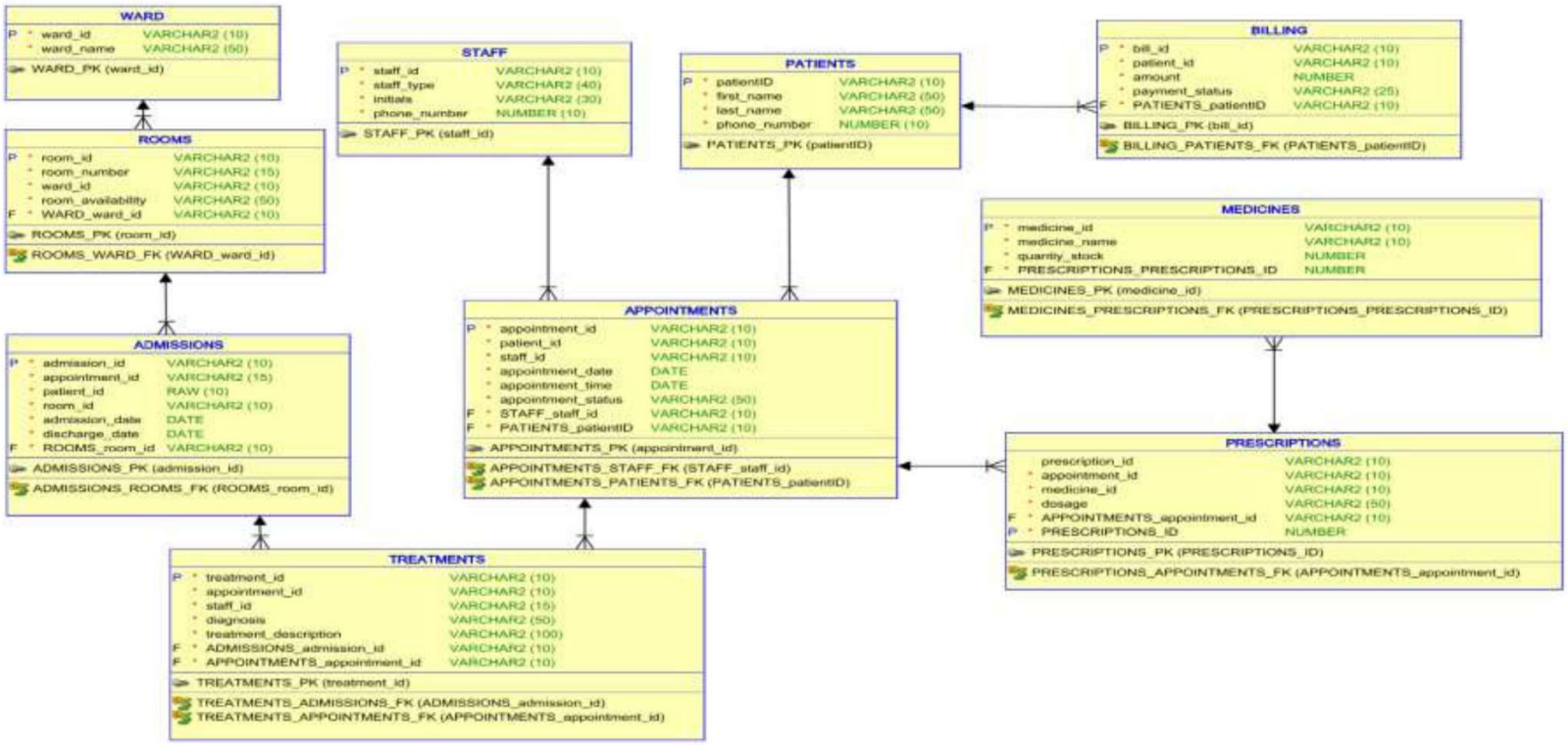
With the G14 database normalized, we can move to implementation or testing of the physical database.

ERD Phase: Planning and Crafting

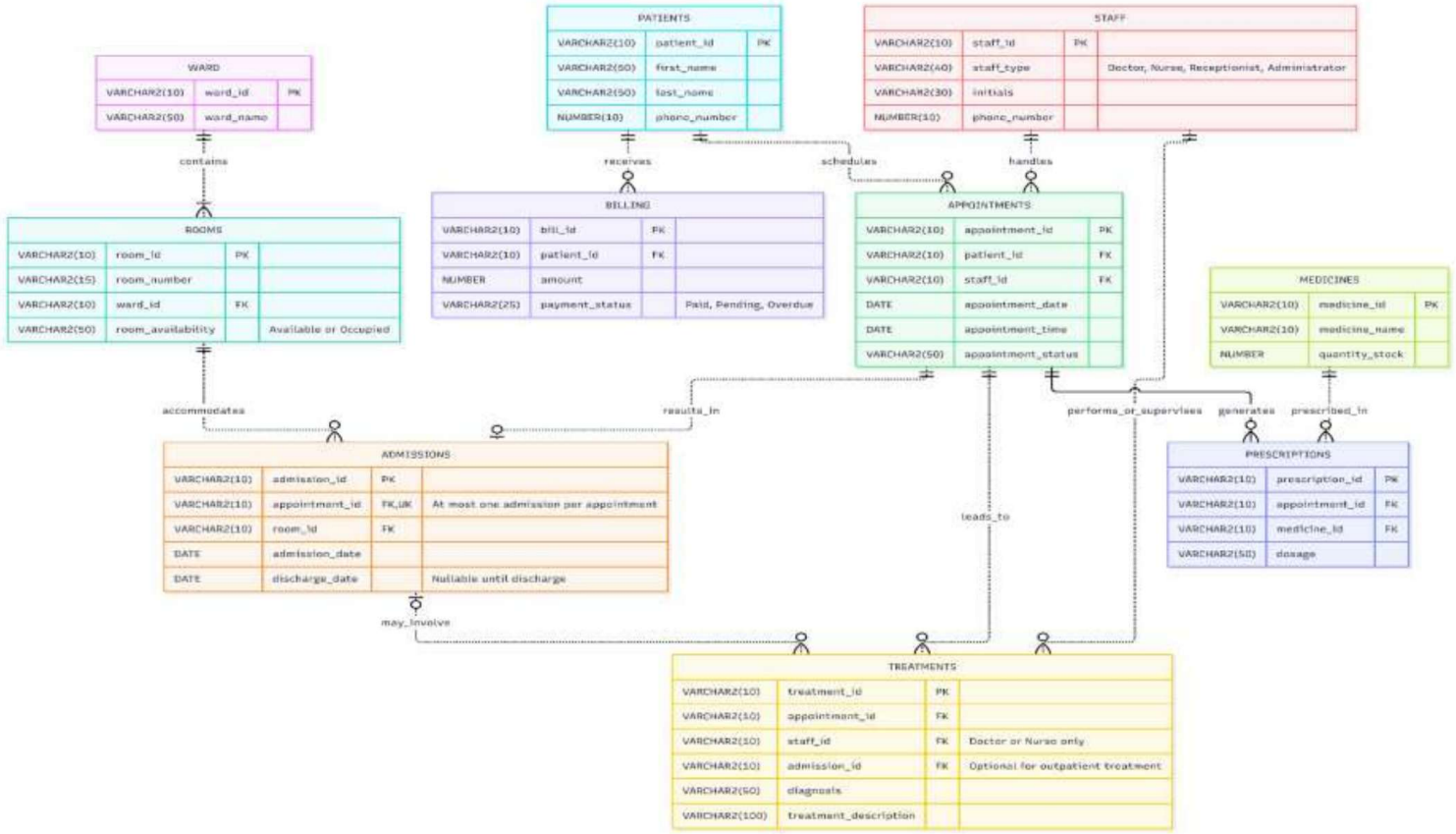
Logical: we used Oracle Data Modeler to design the initial ERDs, both logical and relational.



ERD Phase: Relational



Final ERD (with crow's foot notation)



SQL Code: To Create the Tables

Tables, in creation order

PATIENTS	STAFF
WARD	ROOMS
APPOINTMENTS	ADMISSIONS
TREATMENTS	MEDICINES
PRESCRIPTIONS	BILLING

Example: PATIENTS

```
CREATE TABLE PATIENTS (
  patient_id   VARCHAR(10) NOT NULL,
  first_name   VARCHAR(50) NOT NULL,
  last_name    VARCHAR(50) NOT NULL,
  phone_number VARCHAR(10) NOT NULL,
  CONSTRAINT PATIENTS_PK
    PRIMARY KEY (patient_id)
);
```

SQL Code: Keys and Check Constraints

ROOMS: foreign key and status check

```
CREATE TABLE ROOMS (
  room_id VARCHAR(10) NOT NULL,
  room_number VARCHAR(15) NOT NULL,
  ward_id VARCHAR(10) NOT NULL,
  room_availability VARCHAR(50) NOT NULL,
  CONSTRAINT ROOMS_PK PRIMARY KEY (room_id),
  CONSTRAINT ROOMS_WARD_FK
    FOREIGN KEY (ward_id)
    REFERENCES WARD (ward_id),
  CONSTRAINT ROOM_STATUS_CHK CHECK (
    room_availability IN
    ('Available','Occupied'))
);
```

Check constraints

```
ALTER TABLE STAFF
ADD CONSTRAINT STAFF_TYPE_CHK
CHECK (staff_type IN
  ('Doctor','Nurse',
   'Receptionist','Administrator'));

ALTER TABLE BILLING
ADD CONSTRAINT PAYMENT_STATUS_CHK
CHECK (payment_status IN
  ('Paid','Pending','Overdue'));
```

SQL Code: To Insert Values into Tables

Sample statements

```
INSERT INTO PATIENTS
VALUES ('P001','JOHN','DOE','0720017896');

INSERT INTO WARD VALUES ('W01','General');

INSERT INTO ROOMS
VALUES ('R01','101','W01','Occupied');

INSERT INTO APPOINTMENTS
VALUES ('A001','P001',
  TO_DATE('01-04-2026','DD-MM-YYYY'),
  '08:00','Completed');
```

Sample data loaded

Table	Rows
PATIENTS	9
WARD	3
ROOMS	5
APPOINTMENTS	8
ADMISSIONS	3
MEDICINES	8
PRESCRIPTIONS	6
BILLING	8

Total Number of Registered Patients

SQL

```
SELECT COUNT(*) AS total_patients
FROM PATIENTS;
```

Output (Oracle Live SQL)

TOTAL_PATIENTS
9

Available Rooms

SQL

```
SELECT room_id, room_number
FROM ROOMS
WHERE room_availability = 'Available';
```

Output (Oracle Live SQL)

ROOM_ID	ROOM_NUMBER
R03	102
R05	202

Patients Currently Admitted

SQL

```
SELECT p.patient_id,
       p.first_name,
       p.last_name,
       a.room_id
FROM PATIENTS p
JOIN ADMISSIONS a
ON p.patient_id = a.patient_id;
```

Output (Oracle Live SQL)

PATIENT_ID	FIRST_NAME	LAST_NAME	ROOM_ID
P001	JOHN	DOE	R01
P002	JANE	DOE	R02
P005	WILL	FOSTER	R04

Number of Admissions per Ward

SQL

```
SELECT w.ward_name,
  COUNT(a.admission_id) total_admissions
FROM WARD w
JOIN ROOMS r ON w.ward_id = r.ward_id
JOIN ADMISSIONS a
  ON r.room_id = a.room_id
GROUP BY w.ward_name;
```

Output (Oracle Live SQL)

WARD_NAME	TOTAL_ADMISSIONS
General	1
Surgical	1
ICU	1

Treatments Performed by Each Staff Member

SQL

```
SELECT s.staff_id,
       s.initials,
  COUNT(t.treatment_id) treatments_done
FROM STAFF s
LEFT JOIN TREATMENTS t
ON s.staff_id = t.staff_id
GROUP BY s.staff_id, s.initials;
```

Output (Oracle Live SQL)

STAFF_ID	INITIALS	TREATMENTS_DONE
SD001	DR N NAIDOO	2
SN002	MN.SMITH	0
SR001	L.HOLLY	0
SD002	DR.L.SMITH	0
SA001	N.MOKOENA	0
SN001	L.MAHLANGU	0
SN003	G.OLESENG	0
SA002	K.NAUME	0

Most Prescribed Medicine

SQL

```
SELECT m.medicine_name,
  COUNT(*) times_prescribed
FROM MEDICINES m
JOIN PRESCRIPTIONS p
ON m.medicine_id = p.medicine_id
GROUP BY m.medicine_name
ORDER BY times_prescribed DESC;
```

Output (Oracle Live SQL)

MEDICINE_NAME	TIMES_PRESCRIBED
Aspirin	2
Amoxicillin	1
Metformin	1
Amlodipine	1
Paracetamol	1

Medicines with Low Stock (stock under 100)

SQL

```
SELECT medicine_id,
       medicine_name,
       stock
FROM MEDICINES
WHERE stock < 100;
```

Output (Oracle Live SQL)

MEDICINE_ID	MEDICINE_NAME	STOCK
M006	Panado	50
M007	Grandpa	10
M008	Benelin	80

Total Revenue Generated

SQL

```
SELECT SUM(amount) AS total_revenue
FROM BILLING
WHERE payment_status = 'Paid';
```

Output (Oracle Live SQL)

TOTAL_REVENUE
6500

Overdue Bills

SQL

```
SELECT *
FROM BILLING
WHERE payment_status = 'Overdue';
```

Output (Oracle Live SQL)

BILL_ID	PATIENT_ID	AMOUNT	PAYMENT_STATUS
B007	P008	1250	Overdue

Grouping Appointments by Status

SQL

```
SELECT appointment_status,
       COUNT(*) total
FROM APPOINTMENTS
GROUP BY appointment_status;
```

Output (Oracle Live SQL)

APPOINTMENT_STATUS	TOTAL
Completed	5
Scheduled	2
Cancelled	1

Conclusion

The problem

Manual flat-file record keeping caused data duplication, slow data retrieval, poor organisation and data integrity problems.

The solution

A relational database designed in Third Normal Form (3NF), modelled with an ERD and implemented in SQL.

The result

Sample queries produced meaningful reports: registered patients, room availability, admissions, treatments, medicine stock, billing and revenue.

Conclusion: What G14 Gains

The database offers G14 Healthcare Facility a robust system for handling healthcare records in a unified, secure and efficient way.

- **Improved data integrity**
- **Minimized data redundancy**
- **Quick access to information and reporting**
- **Smoother administrative processes**
- **A foundation for future expansion**
- **Support for both outpatient and inpatient services**

Thank you

Questions?

Group 14 | G14 Hospital Database Management System

